S.P.M College,Udantpuri

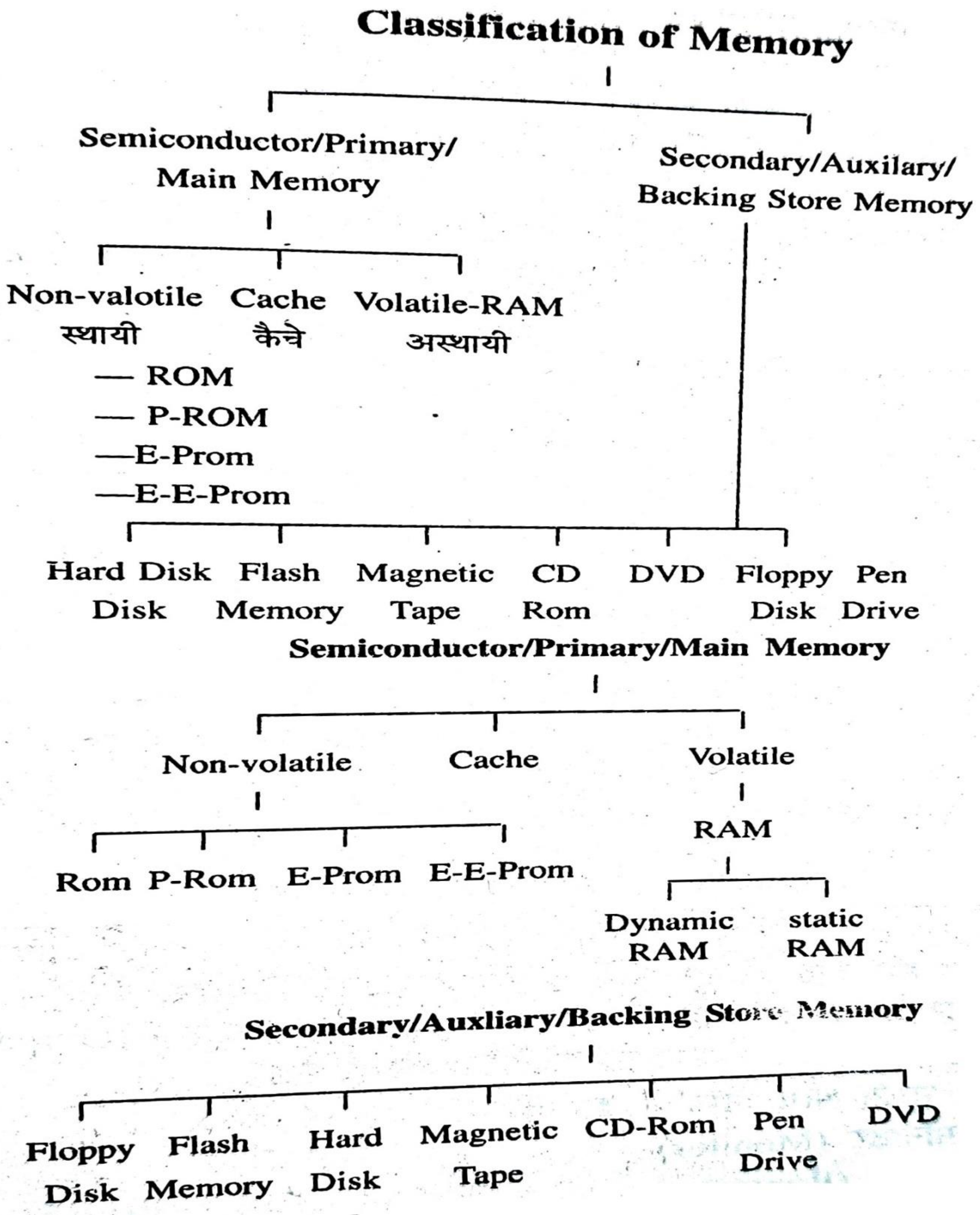
Bachelor Of Computer Application (BCA)

Part -1 (Paper-1) **– Hira Kumar**

Computer Fundamental

STORAGE DEVICE/COMPUTER MEMORY

Memory is that part of the computer in which data & instruction have to store inside the computer, when CPU in processing, this type of device is called Storage Device/Computer Memory.



❖ Primary memory / Main Memory / Internal Memory

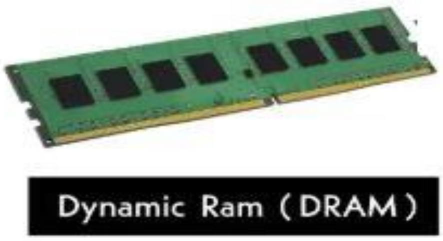
- In primary memory, the programs (or programs) running at any given time and their input and output data are stored for some time.
- As soon as their need is over, they can be removed and other data or programs can be stored.
- It is also called internal memory because it is a part of the computer's CPU.
- This memory is capable of storing & retrieving data very quickly
- The larger the size of the computer's main memory, the faster the computer is considered (माना जाता है) to be.
- Primary memory is only the memory that is directly accessed to the CPU.

Types of Main Memory

1. RAM (Random Access Memory)
2. ROM (Read Only Memory)

1. RAM (Random Access Memory)

- The complete name of RAM is random access memory which is also known as Primary memory.
- It is called read/write memory because data can be read as well as write in RAM.
- All the information present in RAM is temporary and as soon as(जैसे ही) the power supply of the computer is turned off, all the information is destroyed, So it is called volatile memory.



Type of RAM

a)Static RAM b) Dynamic RAM

a) Static RAM

- It is called SRAM.
- Once data is written into the chip, it is maintained until power is supplied to it; it does not need refreshing.
- Faster and consume less power than dynamic RAM.
- It is more expensive. It Uses Transistor.
- Used for CPU cache memory.

What is Cache Memory (कैश मेमोरी) ?

- This is a special type of memory that uses very fast static RAM chips and allows the processor to access a particular memory very quickly.
- Cache Memory acts as a buffer between(b/w) the processor and standard DRAM modules.
- Cache Memory it is also called buffer memory.

b) Dynamic RAM

- It is called DRAM.
- This Storage device use transistor (just like on/off switch) & Capacitors (store electrical charge).
- When transistor on/off capacitors can be charged (1) or uncharge (0).
- It needs refreshing because the charge of the capacitor leaks with time.

Types of DRAM

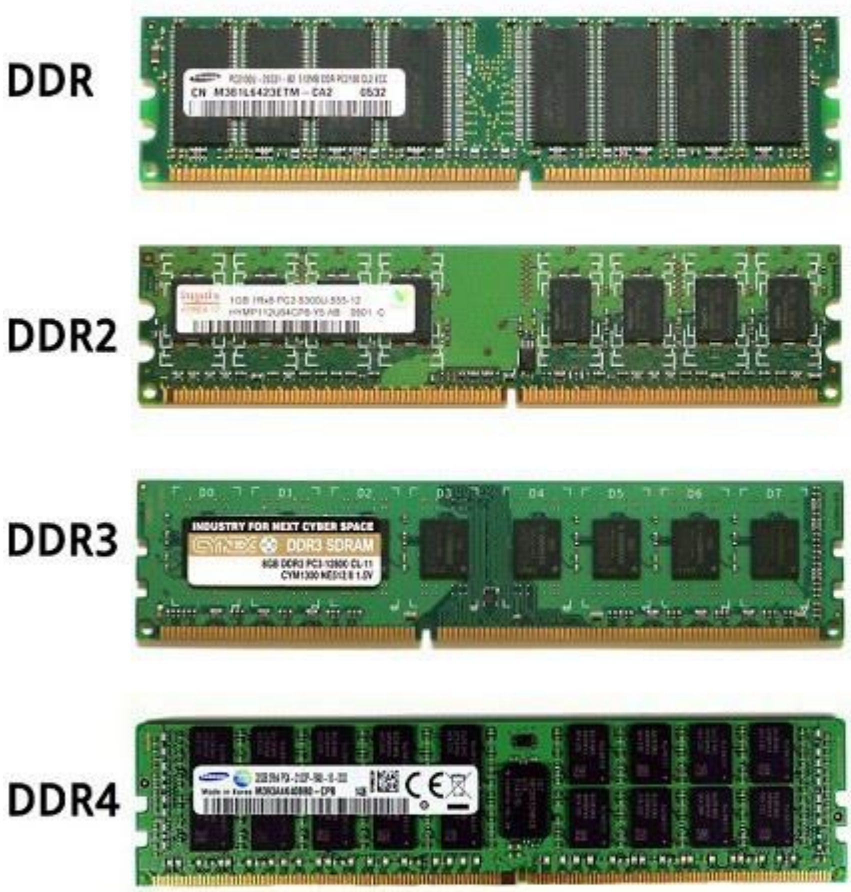
a) SDRAM (Synchronizes Dynamic RAM)

b) RDRAM (Rambus Dynamic RAM)

c) DDRAM (Double Data Dynamic RAM)

Type of DDRAM

i)DDR1 ii)DDR2 (iii) DDR3 (iv) DDR4

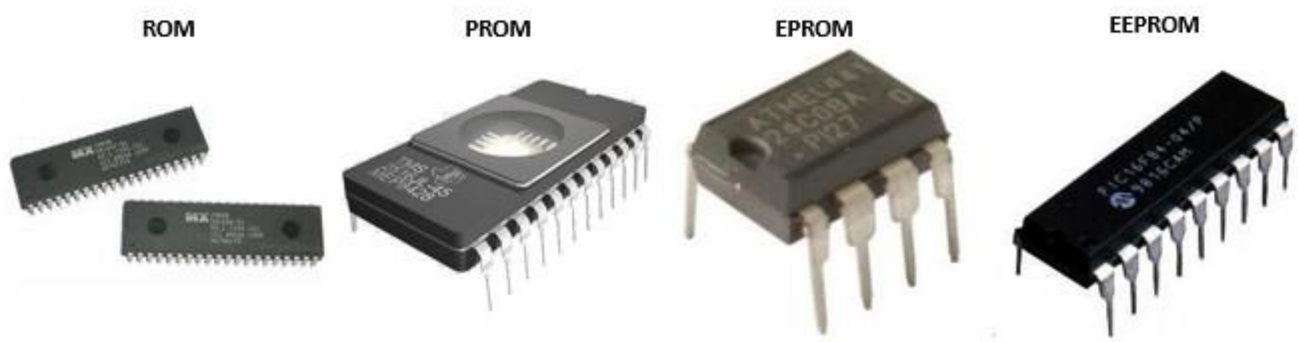


Difference Between DRAM and SRAM

DRAM	SRAM
1. Constructed of tiny capacitors that leak electricity.	1. Constructed of circuits similar to D flip-flops.
2. Requires a recharge every few milliseconds to maintain its data.	2. Holds its contents as long as power is available.
3. Inexpensive.	3. Expensive.
4. Slower than SRAM.	4. Faster than DRAM.
5. Can store many bits per chip.	5. Can not store many bits per chip.
6. Uses less power.	6.Uses more power.
7.Generates less heat.	7.Generates more heat.
8. Used for main memory.	8. Used for cache.

2. ROM (Read Only Memory)

- The complete name of ROM is read only memory.
- The data & instruction are stored permanently & can't be altered/Change by the programmer/user.
- It is a Non – Volatile Memory / Permanent Memory, because the data & instruction are stored permanent,so data is not delete when electric power supply is off.
- This memory also known a field storage, permanent storage or dead storage.
- These programs are stored in ROM. For ex. System Boot Loader.
- Used in Calculators and Peripheral devices.



Types of ROM

i)PROM (Programmable ROM)

- This memory allows the user to program the chip with a PROM writer / Burner Device.
- The chip can be programmed once, there after, it cannot be altered/change.

ii) EPROM (Erasable PROM)

- Permanent Data can be erase by exposing (संपर्क) the memory chip to UV(Ultra Violet) light.
- Chip then can be reprogrammed by electrically.

iii) EEPROM (Electronically Erasable ROM)

- Used to Specific Software / Program Erased and reprogrammed Data through electrically, without removing the chip from the computer.
- Used for storing firmware, BIOS settings etc.

	Cache	RAM	ROM
Stand for	Cache Memory	Randome-Access Memory	Read-Only Memory
Volatility	Volatile	Volatile	Permanent
Function	Enable computers to read data immediately to run applications	Enable computers to read data quickly to run applications	Store the program required to initially boot the computer
Function time	During the normal operations	During the normal operations, if cache is unavailable	When computers start up
Capacity	about 64 KB	up to 16 GB	Usually 4 MB
Static or not	dynamic	dynamic	static
Speed	fastest	faster	slower
Access	Can be read and written; Accessed randomly	Can be read and written; Accessed randomly	can only be read in sequence
Types	Level 1, Level 2, and Level 3	SRAM and PRAM	PROM, EPROM and EEPROM
Location	sometimes inside CPU	closer to CPU	farther from CPU

================HIRA KUMAR=================